WASP-1b

R-1262

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-1_171213` · created 2026-07-31T12:52:43+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458100.6814 +/- 0.0064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.101 +/- 0.029
Transit depth (Rp/Rs)^2	1.01 +/- 0.59 [%]
Orbital Inclination (inc)	86.09 +/- 1.37
Airmass coefficient 1 (a1)	1.2522 +/- 0.0087
Airmass coefficient 2 (a2)	-0.134 +/- 0.0087
Scatter in the residuals of the lightcurve fit is	0.69 %
Best Comparison Star	None
Optimal Aperture	3.94
Optimal Annulus	5.58
Transit Duration (day)	0.0763 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.101 ± 0.029 vs 0.1036 (deviation 0.09σ)
Duration fit vs published	0.0763 d vs 0.1567 d (deviation 10.44σ)
Mid-time O–C	22.0 min
Depth SNR	3.5
Residual scatter	0.69 %

Light curve

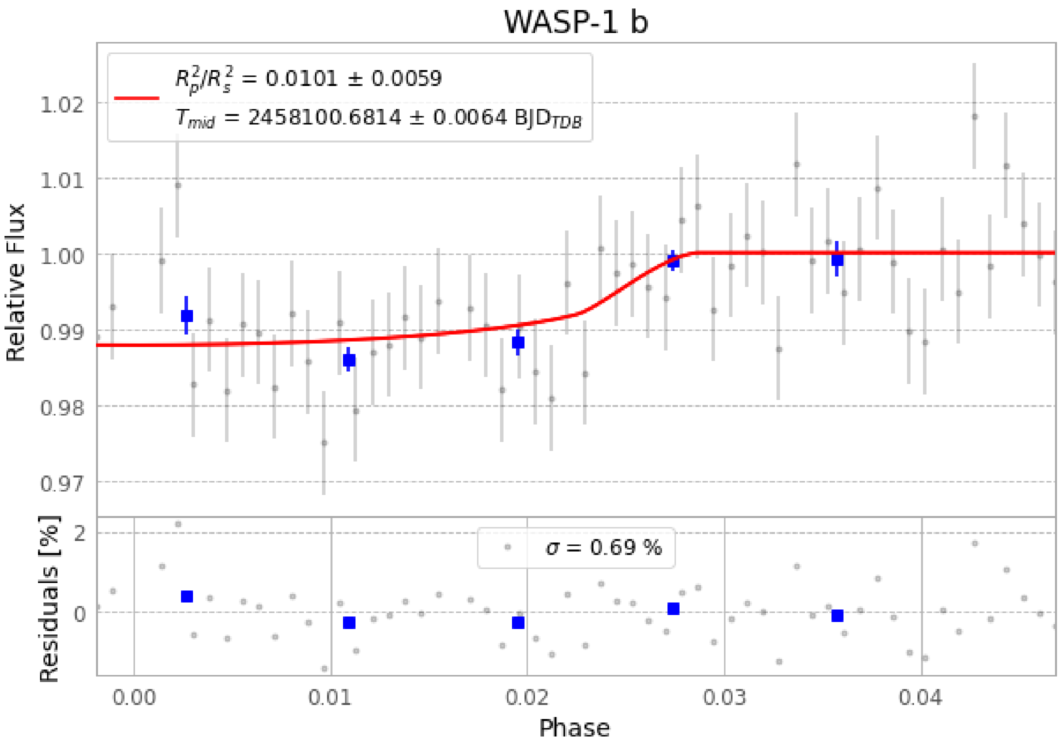

Figure 1 — FinalLightCurve_WASP-1 b_2017-12-12.png (SHA-256 608237248b85bdbf…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [220, 313], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 63154c5ee10260bf…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

58 FITS frames (38 MB), `WASP-11712130409 12.FITS` → `WASP-1171213070612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-171213.log (a652e6c6328b…), FinalLightCurve_WASP-1 b_2017-12-12.png (608237248b85…), FinalLightCurve_WASP-1 b_2017-12-12.csv (8a07de4c5045…)

Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)